INTERNSHIP PROJECT REPORT

Student Performance Analysis and Prediction System

A Python-based Machine Learning project
using Linear Regression and Ridge Regression

Student Name	Amruta Dipak Gaikwad
Roll Number	AIML-A6/SEP-11774
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Abstract

This project analyzes student academic performance and builds machine learning models to estimate a student's final Mathematics grade. The work uses the Student Performance dataset from the UCI Machine Learning Repository. It includes data inspection, exploratory data analysis (EDA), preprocessing, feature engineering, model training, cross-validation, hyperparameter tuning, and evaluation. Linear Regression and Ridge Regression are compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The tuned Ridge Regression model achieved an MAE of 1.589, RMSE of 2.328, and R^2 of 0.736 on the held-out test set. The report also discusses the limitations of using a relatively small, single-subject dataset and the importance of using predictions as decision support rather than as a final judgment about a student.

1. Introduction

Student performance is influenced by several academic, personal, and family-related factors. Understanding patterns in student data can help educators identify trends and provide timely academic support. This project applies data analysis and supervised machine learning to study the factors associated with final Mathematics grades and predict final grades from available student attributes.

2. Problem Statement

Educational institutions may have student records but may not have a simple analytical approach to explore performance patterns or estimate final grades. The project addresses this by building a reproducible Python workflow that examines student data and compares regression models for final-grade prediction.

3. Project Objectives

- Understand the structure and quality of the student performance dataset.
- Explore distributions, correlations, and relationships between selected student attributes and final grades.
- Clean and preprocess data for machine learning.
- Train and compare Linear Regression and Ridge Regression models.
- Evaluate performance using MAE, RMSE, R^2, and cross-validation.
- Tune Ridge Regression and interpret the results.
- Present visualizations and discuss limitations and future improvements.

4. Dataset Description and Source

The project uses the Mathematics subset (student-mat.csv) of the UCI Student Performance dataset. It contains 395 student records and 33 original columns. The attributes include demographic information, family and social background, study habits, school-related information, absences, and grades. The target variable is G3, representing the final Mathematics grade on a 0–20 scale.

Dataset property	Description
Dataset name	Student Performance – Mathematics (student-mat.csv)
Number of records	395
Original columns	33
Prediction target	G3 – final Mathematics grade (0–20)
Data source	UCI Machine Learning Repository
Dataset URL	https://archive.ics.uci.edu/dataset/320/student+performance

The dataset is provided as a semicolon-separated CSV file. It is a public educational dataset and is used here for academic analysis.

5. Data Preprocessing and Feature Engineering

The CSV file was loaded using pandas. Initial inspection included checking the dataset shape, column names, data types, missing values, duplicate rows, and descriptive statistics. The inspection found 0 missing values and 0 duplicate rows in the 395 records.

Preprocessing steps

• Separated the target variable G3 from the input features.
• Excluded G3 from the input data because it is the value being predicted.
• Excluded the derived Total_Grades feature from model inputs because it includes G3 and would cause target leakage.
• Used preprocessing pipelines to handle numerical and categorical features appropriately.
• Applied one-hot encoding to categorical variables and scaling to numerical variables as part of the model workflow.
• Split the data into training and testing subsets using an 80:20 split (316 training records and 79 testing records).

Feature engineering

A Total_Grades variable was created for exploratory analysis by combining the period-grade columns G1, G2, and G3. This feature was used only for analysis and was not supplied to the prediction models because it contains the target grade G3.

6. Exploratory Data Analysis and Visualizations

EDA was performed to understand grade distribution and explore relationships between student attributes and final grades. The final-grade mean was 10.42, the median was 11.00, and the standard deviation was 4.58. The observed correlation between study time and final grade was approximately 0.098, while the correlation between absences and final grade was approximately 0.034 in this dataset. These values indicate weak linear relationships when each factor is considered alone; they do not establish causation.

EDA Visualizations (1–2)

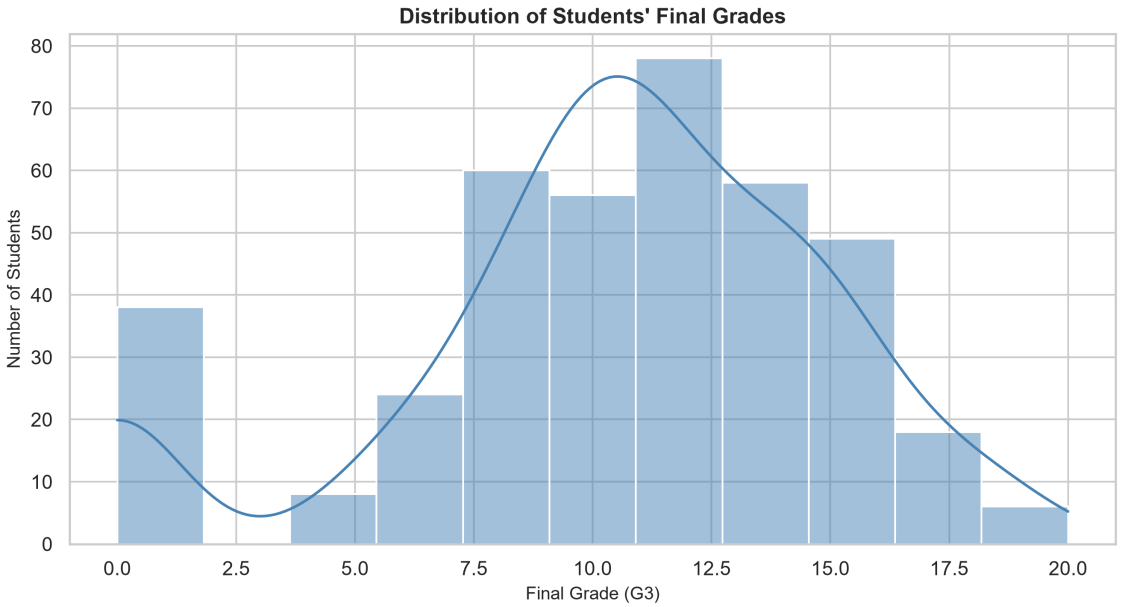


Figure 1. Distribution of students' final Mathematics grades.

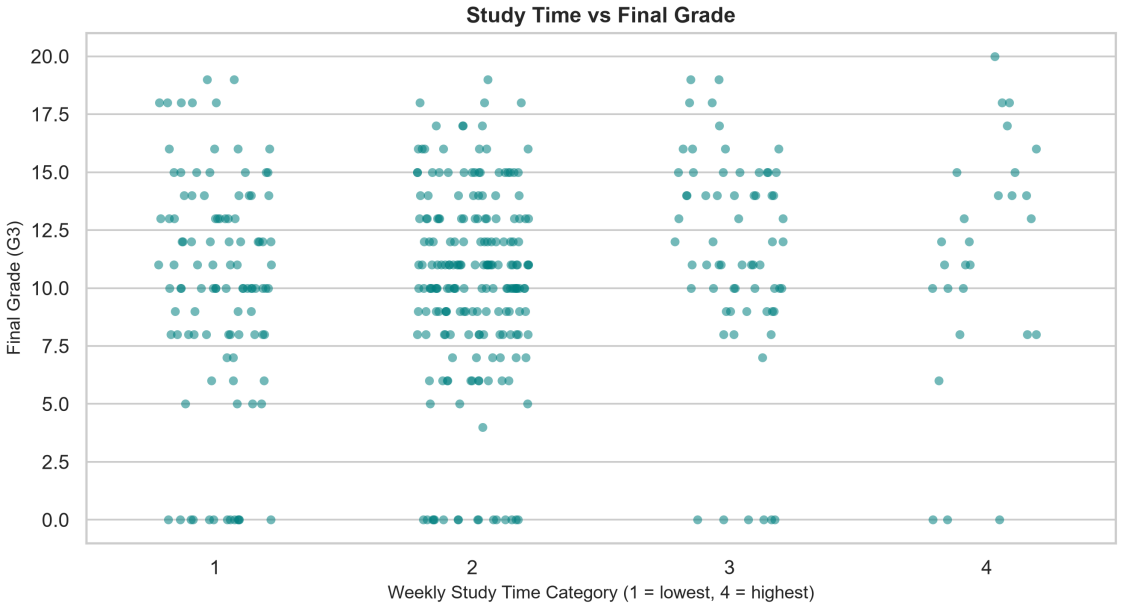


Figure 2. Study time compared with final Mathematics grade.

EDA Visualizations (3–4)

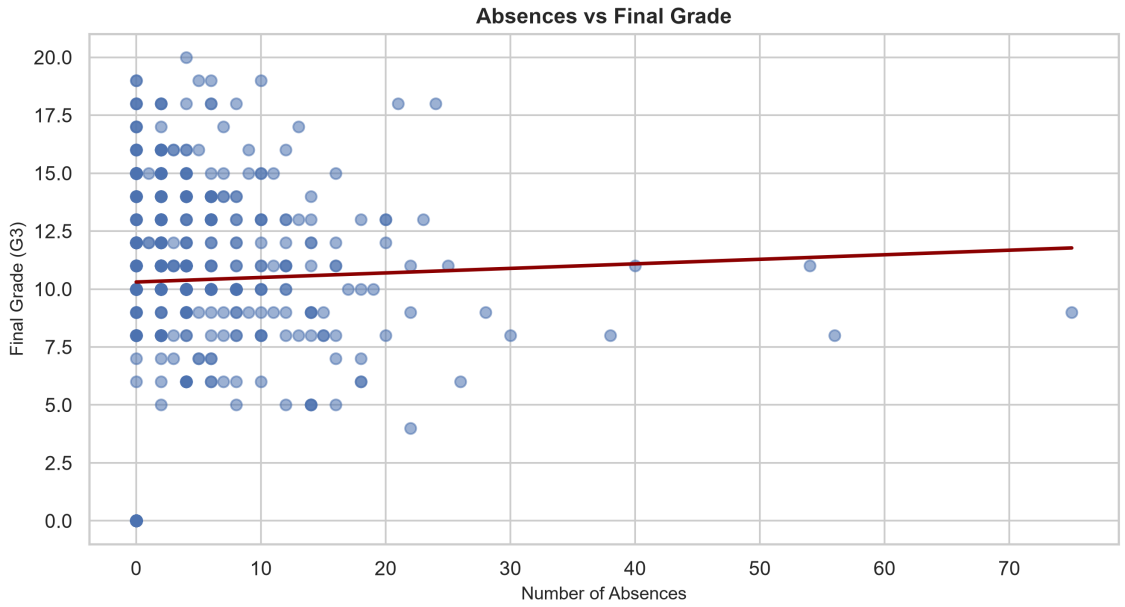


Figure 3. Absences compared with final Mathematics grade.

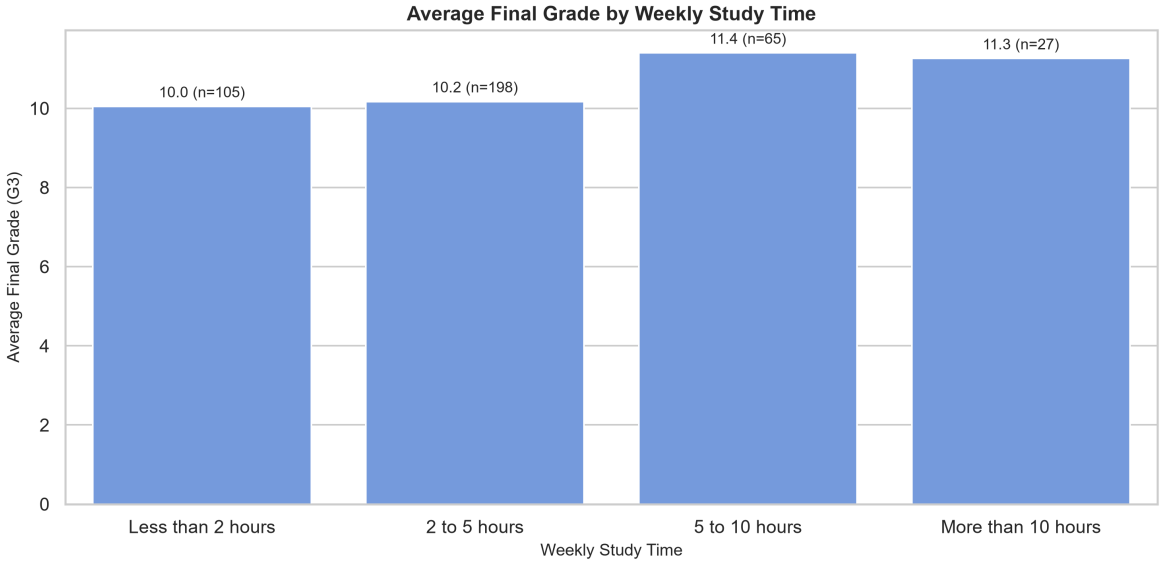


Figure 4. Average final grade across study-time categories.

EDA Visualizations (5–6)

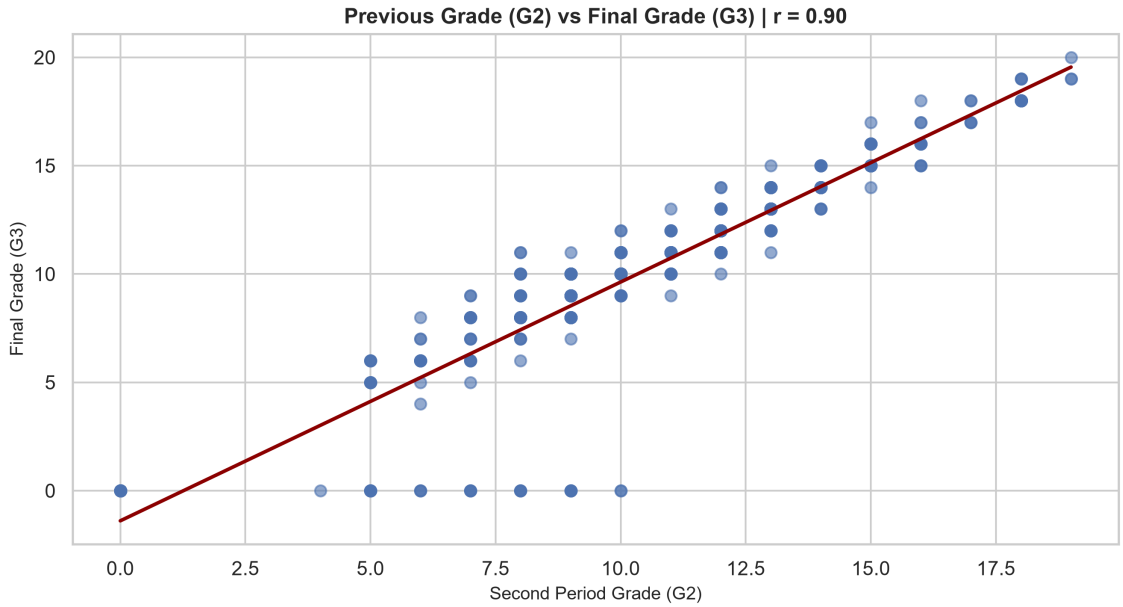


Figure 5. Relationship between previous-period grade (G2) and final grade (G3).

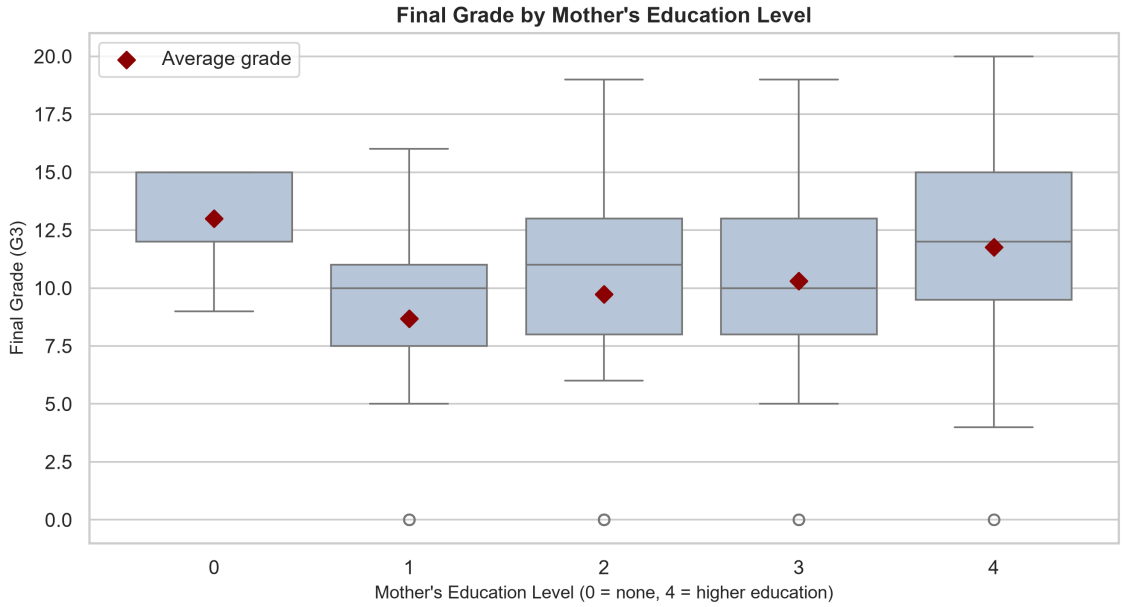


Figure 6. Final grade grouped by mother's education level.

EDA Visualizations (7–8)

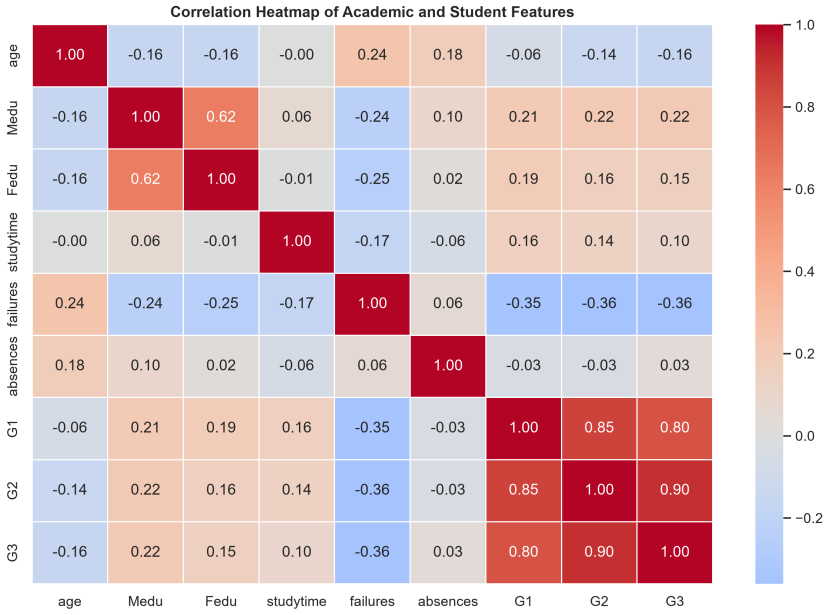


Figure 7. Correlation heatmap for numerical variables.

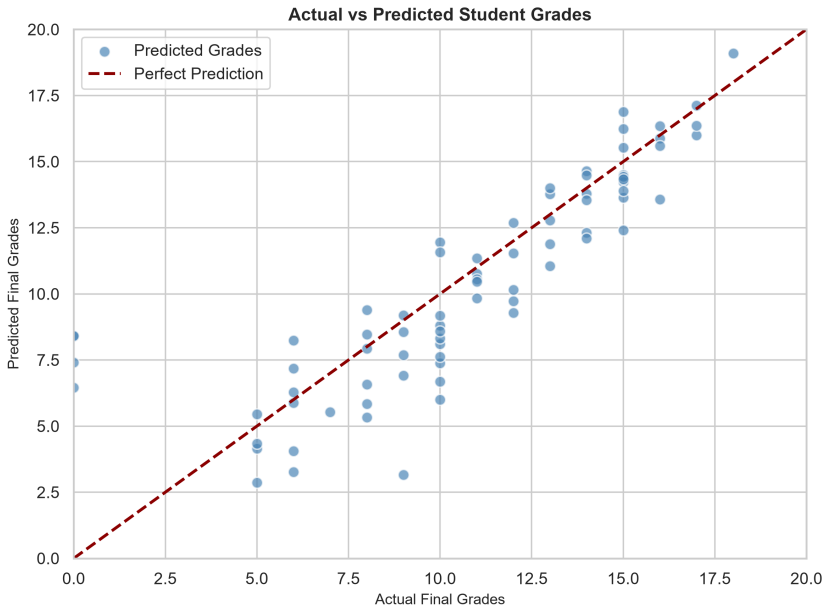


Figure 8. Actual grades compared with model-predicted grades.

EDA Visualizations (9–9)

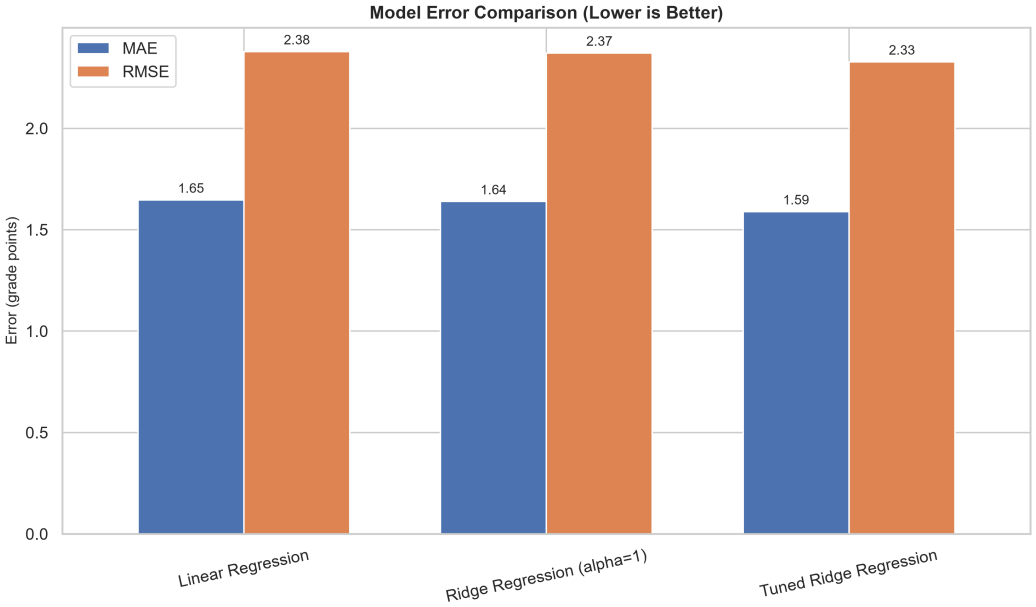


Figure 9. Comparison of model evaluation errors.

7. Machine Learning Methodology

This is a supervised regression problem because the target G3 is a numerical grade. Two regression algorithms were trained and evaluated: Linear Regression and Ridge Regression. Both were implemented using scikit-learn pipelines so that preprocessing is fitted on the training data and then applied consistently to the test data.

Linear Regression

Linear Regression estimates a linear relationship between input features and the target grade. It is used as a baseline model because it is simple and interpretable.

Ridge Regression

Ridge Regression is a regularized form of linear regression. It adds an L2 penalty to the loss function to reduce the size of coefficients and can help control overfitting, particularly when input features are correlated.

Training and validation approach

The dataset was divided into 316 training records and 79 testing records. Five-fold cross-validation was used to estimate the stability of the Linear Regression model. Ridge Regression hyperparameter tuning was performed with GridSearchCV over alpha values [0.1, 1, 10, 100]. The best alpha found was 10, with a best cross-validation R^2 of 0.83.

8. Evaluation Metrics

Metric	Meaning
MAE	Mean absolute difference between actual and predicted grades. Lower is better.
RMSE	Square root of the mean squared prediction error. It penalizes larger errors more strongly. Lower is better.
R^2	Proportion of target variance explained by the model on the evaluated data. Values closer to 1 indicate a better fit.

9. Model Results

Model	MAE	RMSE	R^2
Linear Regression	1.647	2.378	0.724
Ridge Regression (alpha=1)	1.639	2.372	0.726
Tuned Ridge Regression (alpha=10)	1.589	2.328	0.736

The tuned Ridge Regression model produced the best test-set results among the compared models. Its R^2 of 0.736 indicates that it explains approximately 73.6% of the variation in final grades in this test split. Its MAE of 1.589 means the average absolute prediction error was about 1.59 grade points on the 0–20 scale.

Cross-validation results for Linear Regression

The five fold R^2 scores were 0.801, 0.883, 0.768, 0.774, and 0.696. The mean was 0.784 and the standard deviation was 0.061. The variation across folds shows that performance can differ depending on which records are held out for validation.

10. Actual vs. Predicted Grades

The project generated an actual-versus-predicted comparison for the test records and saved the comparison as a visualization and output file. A few example predictions are shown below. These examples illustrate that individual predictions can differ from the true grade, even when the overall model metrics are reasonable.

Actual grade	Predicted grade
10	6.00
12	11.53
5	2.87
10	8.80
9	8.55
13	11.89
18	19.10
6	7.17
0	7.41
14	12.31

11. Interpretation and Findings

• The dataset contains 395 records and has no missing values or duplicate rows in the inspected file.
• The average final Mathematics grade is 10.42 out of 20, with a median of 11.
• Study time alone has a weak positive linear correlation with the final grade (approximately 0.098).
• Absences alone have a weak linear correlation with the final grade (approximately 0.034).
• Previous-period grade information is included in the dataset and is useful for understanding performance patterns; however, the model's input configuration must continue to exclude the target G3 itself.
• Tuned Ridge Regression performed best among the three reported model configurations, with MAE 1.589, RMSE 2.328, and R^2 0.736.

12. Limitations

• The dataset contains only 395 Mathematics records, so results may not generalize to larger or different student populations.
• The data comes from a specific educational context and may not represent all schools or regions.
• The available attendance-related field is the number of absences, not a complete attendance percentage or daily attendance history.
• Correlation does not imply causation; the analysis cannot prove that a particular factor causes a change in grades.
• The model predicts grades from recorded attributes and should not replace teacher judgment or student support conversations.
• Performance is evaluated on one train/test split; further validation on independent datasets would improve confidence.

13. Future Improvements

• Collect a larger and more diverse dataset from multiple schools and academic years.
• Include richer attendance, assessment, and learning-engagement information where permitted.
• Evaluate additional regression models such as Random Forest Regressor, Gradient Boosting, and XGBoost using a consistent validation strategy.
• Perform error analysis across student groups to identify systematic prediction differences.
• Add explainability methods such as permutation importance or SHAP to better understand model behavior.
• Build a simple dashboard for educators to explore aggregate trends while protecting student privacy.

14. Conclusion

The Student Performance Analysis and Prediction System demonstrates an end-to-end machine learning workflow using Python. The project inspected and visualized the UCI Student Performance dataset, applied preprocessing and feature engineering, and compared Linear Regression with Ridge Regression. Tuned Ridge Regression achieved the strongest reported test performance, with MAE 1.589, RMSE 2.328, and R^2 0.736. The analysis also highlighted that individual factors such as study time and absences have weak standalone linear correlations with final grades in this dataset. The project provides a useful academic demonstration of regression-based analysis, while its limitations show why predictions should be interpreted carefully and validated with broader data.

15. References

1. Cortez, P. and Silva, A. (2008). Using data mining to predict secondary school student performance. In Proceedings of the 5th Future Business Technology Conference (FUBUTEC).
2. UCI Machine Learning Repository. Student Performance dataset. https://archive.ics.uci.edu/dataset/320/student+performance
3. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
4. pandas documentation: https://pandas.pydata.org/docs/
5. Matplotlib documentation: https://matplotlib.org/stable/
6. Seaborn documentation: https://seaborn.pydata.org/

End of Report